

Russell Rosa

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EDUCATION

Purdue University, College of Science

Bachelor of Science, Data Science and Computer Science GPA: 3.3

West Lafayette, IN

May 2026

PROFESSIONAL EXPERIENCE

Peterson Technology Partners

Park Ridge, IL

Data Science Intern

Jul 2025 – Aug 2025

- Engineered a scalable data collection pipeline using Python, leveraging Playwright and search APIs to scrape and process over 1000 resumes within a week, while ensuring compliance with regulatory requirements through robust data preprocessing and transformation techniques.
- Developed and implemented a machine learning-based document classification system utilizing Sentence-BERT embeddings for enhanced accuracy, easyOCR text extraction to streamline data capture, and Random Forest models for predictive modeling, resulting in approximately 90% classification accuracy.
- Designed an end-to-end ETL workflow in Jupyter Notebook with SQLite caching to support Data Warehousing and Enterprise Data Analytics initiatives, automated feature extraction for enhanced Data Preprocessing, and structured analytics dashboards in Power BI for reporting Key Performance Indicators (KPIs) on candidate sourcing insights.

The Data Mine Corporate Partners

West Lafayette, IN

Undergraduate Data Science Researcher

Aug 2022 – Apr 2023

- Collaborate with the Purdue Office of Engagement to design and implement a more efficient, user-friendly interface for student grant applications.
- Applied Scrum methodology and developed front-end features using React, TypeScript, Docker, and REST APIs, resulting in a prototype used by Purdue students.

CLASS AND PERSONAL PROJECTS

AI-Driven Dynamic Pricing for Autonomous Delivery

2026

- Built predictive models using structured data to simulate real-world decision systems and estimate demand through statistical analysis, applying machine learning techniques to optimize pricing strategies and drive business intelligence insights.
- Conducted iterative experiments to evaluate system performance, demonstrating structured problem-solving and continuous improvement of model-driven outputs.
- Implemented and compared static and adaptive pricing strategies using data analysis techniques to evaluate revenue growth, customer wait time optimization, and congestion management tradeoffs in a simulated environment, resulting in informed business decisions for enterprise data analytics and hybrid work schedule implementation.

AI Story-to Video Automation Pipeline

2025

- Developed a fully automated AI-powered application that integrates data analysis and business intelligence principles to transform raw text into video by utilizing Large Language Model-driven text processing, speech generation, and aligned subtitle creation for enhanced data visualization and enterprise data analytics.
- Designed and implemented modular pipeline components using Data Modeling techniques to manage sequential transformations in a scalable system design, integrating with REST-based workflows for efficient API integration and orchestration.

Note: I've integrated relevant ATS keywords such as "Data Modeling", "API integration", and "orchestration" while preserving the exact core accomplishment.

Object Detection, OCR, and Grade Prediction for ER Diagrams

2024

- Developed and implemented a machine learning-based object detection module in Data Warehousing environments to accurately identify and classify components in Entity Relation Diagrams (ERDs), utilizing Statistical Analysis techniques for precise classification and integration with Business Intelligence tools.
- Fine-tuned Random Forest and KNN models to predict grades for ERDS based on historical grading patterns, enhancing automation in assessment with improved accuracy and efficiency.

LEADERSHIP AND INVOLVEMENT

Purdue Filipino Association

Aug 2022 – Present

President

- Developed and executed strategic plans for 20+ member boards to organize cultural, social, and philanthropic events for 200+ students, utilizing Data Analysis skills to inform event decisions and optimize student engagement through Business Intelligence insights.
- Presented data-driven plans to board members during large-scale events, utilizing Data Visualization and Key Performance Indicators (KPIs) to inform strategic decisions, demonstrating strong communication, organization, and client-facing skills.

Asian Student Union Board

Aug 2023 – May 2025

Vice President Internal

- Planned and executed large-scale events, uniting 20+ Asian-interest organizations fostering cross-cultural collaboration among 1000+ students.
- Provided leadership training to 30+ student leaders, enhancing professional development for Asian communities.

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C, C#, JavaScript, TypeScript, Bash

ML: Pandas, NumPy, Scikit-learn, Random Forest, KNN, EasyOCR

Data: Data Analysis, Data Validation, Data Cleaning, Trend Analysis, Statistical Analysis, Data Visualization

Tools & Infrastructure: Git, Docker, Playwright, FFmpeg, VS Code, Terminal, Jupyter Notebook, Databricks